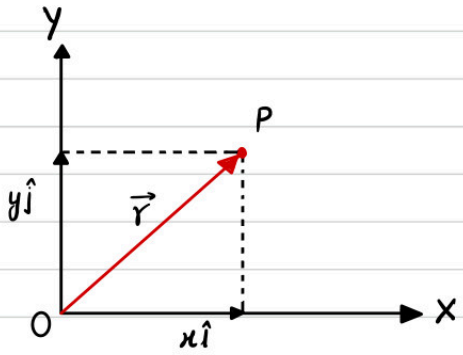
Motion in plane

When a particle moves in two dimensions, its motion is called two-dimensional motion or motion in plane.

Position vector in a plane:-

A vector that expresses the position of any particle with respect to a reference point or the origin.



Acc. to diagram position vector of point P

in 2-D

$$\vec{r} = \vec{OP}$$

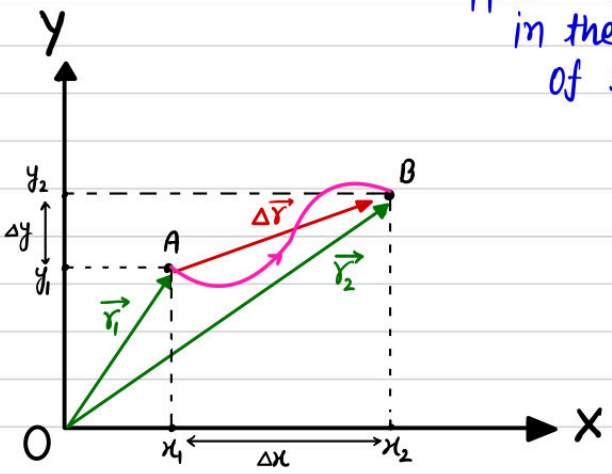
$$\boxed{\vec{r} = x\hat{i} + y\hat{j}}$$

In 3-D space

$$\boxed{\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}}$$

Displacement vector:-

Suppose a particle is displaced from point A to B in the X-Y plane. Then, displacement vector of the particle is



$$\vec{AB} = \vec{OB} - \vec{OA}$$

$$\Delta\vec{r} = \vec{r_2} - \vec{r_1}$$

$$= (x_2\hat{i} + y_2\hat{j}) - (x_1\hat{i} + y_1\hat{j})$$

$$= (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j}$$

$$\boxed{\Delta\vec{r} = \Delta x\hat{i} + \Delta y\hat{j}}$$

Velocity of a particle in a plane:-

$$\text{Average velocity} = \frac{\text{Total displacement}}{\text{time taken}}$$

$$\vec{V_{av}} = \frac{\Delta\vec{r}}{\Delta t} = \frac{\Delta x\hat{i} + \Delta y\hat{j}}{\Delta t}$$

$$= \left(\frac{\Delta x}{\Delta t}\right)\hat{i} + \left(\frac{\Delta y}{\Delta t}\right)\hat{j}$$

$$\boxed{\vec{V_{av}} = V_x\hat{i} + V_y\hat{j}}$$

Magnitude

$$V_{av} = \sqrt{V_x^2 + V_y^2}$$

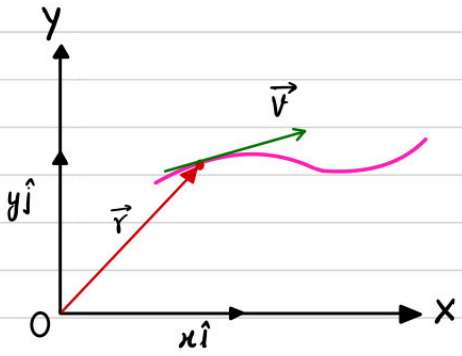
direction

$$\boxed{\tan\theta = \frac{V_y}{V_x}}$$

Instantaneous velocity

$$\vec{V} = \lim_{\Delta t \to 0}\left(\frac{\Delta \vec{r}}{\Delta t}\right)$$

$$\boxed{\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j}}$$

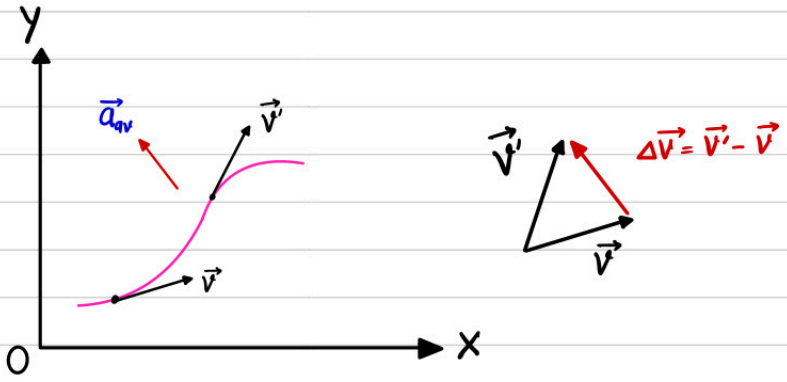


Average acceleration :-

$$\vec{a}_{av} = \frac{\Delta \vec{V}}{\Delta t} = \frac{\Delta V_x \hat{i} + \Delta V_y \hat{j}}{\Delta t}$$

$$= \left(\frac{\Delta V_x}{\Delta t}\right)\hat{i} + \left(\frac{\Delta V_y}{\Delta t}\right)\hat{j}$$

$$\boxed{\vec{a}_{av} = a_x \hat{i} + a_y \hat{j}}$$



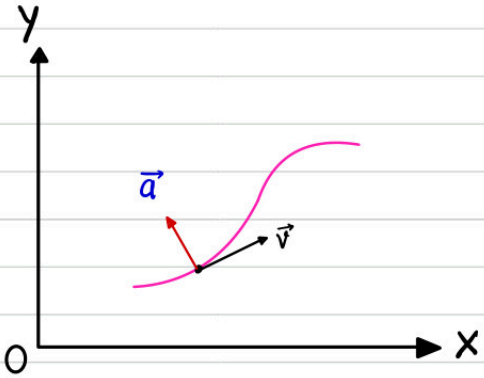
Instantaneous acceleration :-

$$\vec{a} = \lim_{\Delta t \to 0}\left(\frac{\Delta \vec{v}}{\Delta t}\right)$$

$$\boxed{\vec{a} = \frac{d\vec{v}}{dt} = \frac{d(v_x)}{dt}\hat{i} + \frac{d(v_y)}{dt}\hat{j}}$$

$$\boxed{\vec{a} = a_x \hat{i} + a_y \hat{j}}$$

$$\boxed{a = \sqrt{a_x^2 + a_y^2}}$$



Example 4.4 The position of a particle is given by

$$\mathbf{r} = 3.0t\,\hat{\mathbf{i}} + 2.0t^2\hat{\mathbf{j}} + 5.0\hat{\mathbf{k}}$$

where t is in seconds and the coefficients have the proper units for $\mathbf{r}$ to be in metres. (a) Find $\mathbf{v}(t)$ and $\mathbf{a}(t)$ of the particle. (b) Find the magnitude and direction of $\mathbf{v}(t)$ at $t = 1.0$ s.

Sol:-

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(3t\,\hat{i} + 2t^2\hat{j} + 5\hat{k})$$

$$\vec{v} = 3\hat{i} + 4t\hat{j}$$

acceleration $\vec{a} = \frac{d\vec{v}}{dt}$

$\vec{a} = 4\hat{j}$ constant

at $t = 1$ sec

Velocity $\vec{v} = 3\hat{i} + 4\hat{j}$

magnitude $V = \sqrt{3^2 + 4^2} = 5$ m/s

direction $\theta = \tan^{-1}\left(\frac{v_y}{v_x}\right) = \tan^{-1}\left(\frac{4}{3}\right)$

$\theta = 53^\circ$

Equation of motion in a plane ($\vec{a}$ = constant)

① $\vec{v} = \vec{u} + \vec{a}t$

② $\Delta\vec{r} = \vec{u}t + \frac{1}{2}\vec{a}t^2$

③ $v^2 = u^2 + 2\vec{a}\cdot\Delta\vec{r}$

For X- axis

① $\vec{v_x} = \vec{u_x} + \vec{a_x}t$

② $\Delta\vec{x} = \vec{u_x}t + \frac{1}{2}\vec{a_x}t^2$

③ $v_x^2 = u_x^2 + 2\vec{a_x}\cdot\Delta\vec{x}$

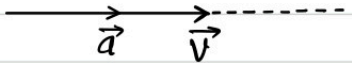
For Y- axis

① $\vec{v_y} = \vec{u_y} + \vec{a_y}t$

② $\Delta\vec{y} = \vec{u_y}t + \frac{1}{2}\vec{a_y}t^2$

③ $v_y^2 = u_y^2 + 2\vec{a_y}\cdot\Delta\vec{y}$

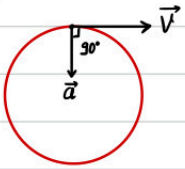
① Straight line Path →
if $\vec{v}$ and $\vec{a}$ parallel or anti parallel



② Curved path →
If angle b/w $\vec{v}$ and $\vec{a}$ is $\theta \neq 0^\circ$ or 180°



③ Circular path →
If angle b/w $\vec{v}$ and $\vec{a}$ is $\theta = 90^\circ$



Example 4.5 A particle starts from origin at $t = 0$ with a velocity $5.0\,\hat{i}$ m/s and moves in x-y plane under action of a force which produces a constant acceleration of $(3.0\hat{i}+2.0\hat{j})$ m/s^2. (a) What is the y-coordinate of the particle at the instant its x-coordinate is 84 m ? (b) What is the speed of the particle at this time ?

Sol:- initial velocity $\vec{u} = 5\hat{i}$ m/s

Here $u_x = 5$ m/s, $u_y = 0$ m/s

acce. $\vec{a} = (3\hat{i} + 2\hat{j})$ m/s^2

hence $a_x = 3$ m/s^2, $a_y = 2$ m/s^2

For X- direction

$x = u_x t + \frac{1}{2}a_x t^2$

$84 = 5\times t + \frac{1}{2}\times 3 \times t^2$

$\Rightarrow 1.5t^2 + 5t - 84 = 0$

hence $t = 6$ sec

For Y- direction

$y = u_y t + \frac{1}{2}a_y t^2$

$y = 0\times 6 + \frac{1}{2}\times 2 \times 36 = 36$ m

after 6 second

$\vec{v_x} = \vec{u_x} + \vec{a_x}t$

$= 5\hat{i} + 3\times 6\,\hat{i}$

$= 23\hat{i}$

and $\vec{v_y} = \vec{u_y} + \vec{a_y}t$

$\vec{v_y} = 0 + 2\times 6\hat{j} = 12\hat{j}$

resultant velocity

$\vec{v} = 23\hat{i} + 12\hat{j}$

speed $= \sqrt{23^2 + 12^2} \simeq 26$ m/s

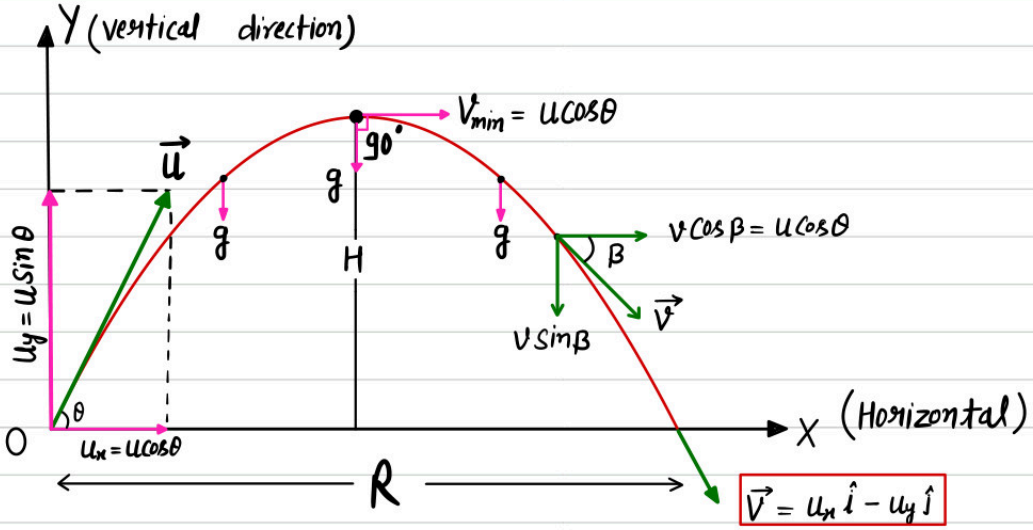
Projectile motion:- If an object is thrown upwards from the Earth's surface at an angle, and its moves only under the influence of gravitational force, then this is called projectile motion. A body undergoing projectile motion is called projectile.

Example :-
① The motion of a bullet fired from a rifle.
② The motion of a ball when a sixt is hit in a cricket match.

Assumptions before studying projectile motion:-

① Air friction is considered negligible.
② The magnitude and direction of gravitational acceleration are considered constant.

Mathematical analysis of projectile motion:-



Let a particle be projected with an initial velocity $\vec{u}$ at an angle θ with the horizontal.

Quantity	Horizontal	Vertical	Resultant
Initial velocity	$\vec{u}_x = u\cos\theta\,\hat{i}$	$\vec{u}_y = u\sin\theta\,\hat{j}$	$\vec{u} = u\cos\theta\hat{i} + u\sin\theta\hat{j}$
Acceleration	$\vec{a}_x = 0$	$\vec{a}_y = -g\hat{j}$	$\vec{a} = -g\hat{j}$
Velocity after time t	$\vec{v}_x = \vec{u}_x + \vec{a}_x t$ $\vec{v}_x = \vec{u}_x$	$\vec{v}_y = (u\sin\theta - gt)\hat{j}$	$\vec{v} = u\cos\theta\hat{i} + (u\sin\theta - gt)\hat{j}$
Velocity at highest point	$\vec{v}_x = u\cos\theta\,\hat{i}$	$\vec{v}_y = 0$	$\vec{v}_{min} = u\cos\theta\,\hat{i}$
Final velocity (at $t=T$)	$\vec{u}_x = u\cos\theta\,\hat{i}$	$\vec{u}_y = -u\sin\theta\,\hat{j}$	$\vec{u} = u\cos\theta\hat{i} - u\sin\theta\hat{j}$

Time of ascent:-

For Y-direction $\quad \cancel{v_y}^{0} = u_y + a_y t$

$$0 = u_y - gt \Rightarrow \boxed{t = \frac{u_y}{g} = \frac{u\sin\theta}{g}}$$

Time of flight (T):- The total time for which the projectile remains in air.

$$T = \frac{u_y}{g} + \frac{u_y}{g} \qquad \boxed{T = \frac{2u_y}{g} = \frac{2u\sin\theta}{g}}$$

Maximum height (H):- The maximum vertical distance attained by the projectile.

For Y-direction

$$\cancel{v_y^2}^{0} = u_y^2 + 2a_y \cancel{h}^{H}$$

$$0 = u_y^2 - 2gH$$

$$\Rightarrow \boxed{H = \frac{u_y^2}{2g} = \frac{u^2\sin^2\theta}{2g}}$$

Horizontal range (R):- The horizontal distance covered during the time of flight.

$$R = u_x \times T$$

$$R = u_x \times \frac{2u_y}{T} \Rightarrow \boxed{R = \frac{2u_x u_y}{T}}$$

$$R = \frac{2u\cos\theta \cdot u\sin\theta}{g} = \frac{u^2 2\sin\theta\cdot\cos\theta}{g}$$

$$\boxed{R = \frac{u^2\sin 2\theta}{g}}$$

If $\sin 2\theta = 1 \Rightarrow 2\theta = 90^\circ \Rightarrow \theta = 45^\circ$

Then $R_{max} = \frac{u^2}{g}$ The horizontal range is maximum when $\theta = 45^\circ$

Relation between R and H:-

$$\frac{R}{H} = \frac{2u_x \cancel{u_y}/\cancel{g}}{\cancel{u_y^2}/\cancel{2g}} = \frac{4u_x}{u_y} = \frac{4u\cos\theta}{u\sin\theta} \Rightarrow \boxed{R = 4H\cot\theta} \Rightarrow \boxed{R_{max} = 4H}$$

Ques- If initial velocity of a partical is $\vec{u} = (3\hat{i} + 4\hat{j})$ m/s in projectile motion. Then find out -

(i) Angle of projection from ground
(ii) Horizontal range
(iii) Maximum height
(iv) Time of flight

Sol:-

Given that $\vec{u} = (3\hat{i} + 4\hat{j})$ m/s $\Rightarrow u_x = 3$ m/s and $u_y = 4$ m/s

(i) Angle of Projection

here $u\cos\theta = u_x$

and $u\sin\theta = u_y$

then $\tan\theta = \frac{u_y}{u_x}$

$\tan\theta = \frac{4}{3} \Rightarrow \theta = 53^\circ$

(ii) Horizontal range

$R = \frac{2u_x u_y}{g}$

$R = \frac{2\times3\times4}{10}$

$R = 2.4\ m$

(iii) Maximum height

$H = u_y^2/2g$

$H = \frac{(4)^2}{2\times10}$

$H = \frac{16}{20}$

$H = 0.8\ m$

(iv) Time of flight

$T = 2u_y/g$

$T = \frac{2\times4}{10}$

$T = 0.8$ sec

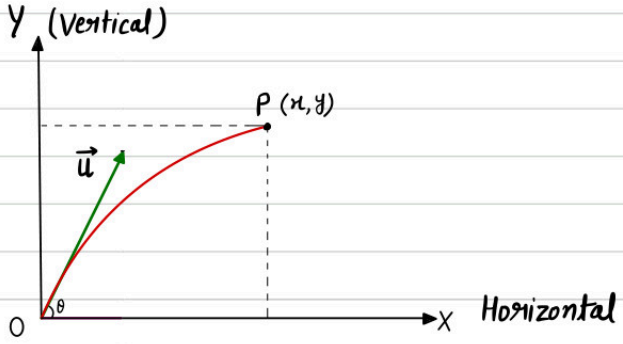
Equation of trajectory:- The path followed by a projectile is called its trajectory.

Let at a time t partical reached at point P.

Then for X-direction

$x = u_x t + \frac{1}{2}\cancel{a_x}^{0} t^2$

$t = \frac{x}{u_x} = \frac{x}{u\cos\theta}$ — (1)



For Y-direction

$y = u_y t - \frac{1}{2} g t^2$

$y = u\sin\theta\left(\frac{x}{u\cos\theta}\right) - \frac{1}{2} g \left(\frac{x}{u\cos\theta}\right)^2$

$$y = (\tan\theta)x - \left(\frac{g}{2u^2\cos^2\theta}\right)x^2 \Rightarrow y = x\tan\theta\left[1 - \frac{x}{R}\right]$$

The path of projectile is parabolic.

$y = ax - bx^2$

$\tan\theta = a$ and $R = \frac{a}{b}$

Remember → Average velocity over the entire flight is $v_{av} = u\cos\theta$

Ques:- The path of a projectile is given by $y = \left(\frac{4}{3}x - \frac{x^2}{18}\right)m$. Then find out angle of projection from ground and horizontal range.

Sol:-

$$y = \frac{4}{3}x - \frac{x^2}{18}$$

$$y = \frac{4}{3}x\left[1 - \frac{x}{24}\right]$$

Compare this equation for $y = x\tan\theta\left[1 - \frac{x}{R}\right]$

$$\tan\theta = \frac{4}{3} \Rightarrow \theta = 53^\circ$$

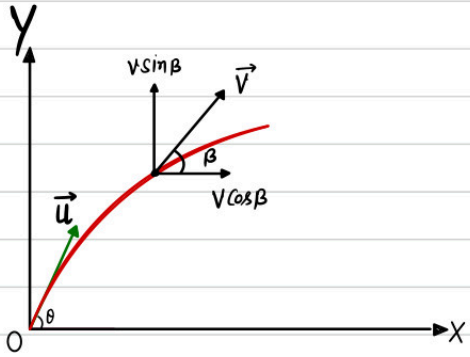
and $R = 24\ m$

Speed of the projectile when it makes an angle β with horizontal:-

The horizontal component of velocity remains constant

$$v\cos\beta = u\cos\theta$$

$$\boxed{v = \frac{u\cos\theta}{\cos\beta}}$$



The time t at which the velocity vector v and initial velocity $\vec{u}$ are perpendicular:-

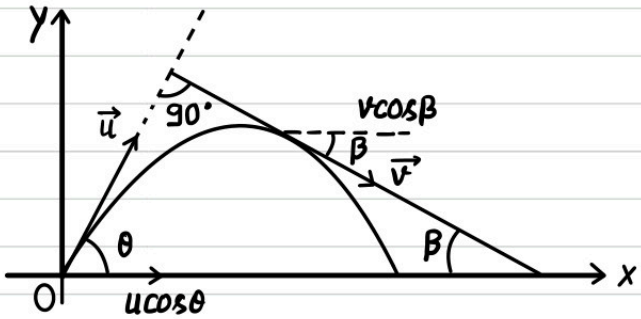
$$\beta = 90^\circ - \theta$$

Here

$$v\cos\beta = u\cos\theta$$

$$v = \frac{u\cos\theta}{\cos(90-\theta)}$$

$$v = \frac{u\cos\theta}{\sin\theta} \Rightarrow \boxed{v = u\cot\theta}$$



When $\vec{v}$ and $\vec{u}$ are perpendicular, their dot product is zero.

$$\vec{u}\cdot\vec{v} = 0$$

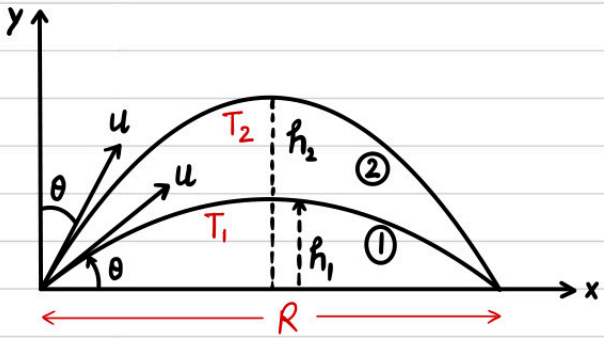
$$(u\cos\theta\,\hat{i} + u\sin\theta\,\hat{j})\cdot[u\cos\theta\,\hat{i} + (u\sin\theta - gt)\hat{j}] = 0$$

$$\Rightarrow u^2\cos^2\theta + u^2\sin^2\theta - u\sin\theta gt = 0$$

$$\Rightarrow u\sin\theta \cdot gt = u^2$$

$$\boxed{t = \frac{u}{g\sin\theta}} \Rightarrow \boxed{t = \frac{u\,\mathrm{cosec}\,\theta}{g}}$$

Projectile for complementary angles:-



For two complementary angles of projection, θ and $(90°-\theta)$, with the same initial speed u.

here $\theta_1 = \theta$ or $(45°-\theta)$,
$\theta_2 = (90°-\theta)$ or $(45°+\theta)$

Then $R_\theta = \frac{u^2 \sin 2\theta}{g}$ — ①

and $R_{(90°-\theta)} = \frac{u^2 \sin 2(90°-\theta)}{g}$

$R_{(90°-\theta)} = \frac{u^2 \sin 2\theta}{g}$ — ②

From equation ① and ②

$$\boxed{R_\theta = R_{(90°-\theta)}}$$

Maximum heights:-

$h_1 = \frac{u^2 \sin^2\theta}{2g}$ — ③

and $h_2 = \frac{u^2 \sin^2(90°-\theta)}{2g} = \frac{u^2 \cos^2\theta}{2g}$ — ④

From equation ③ and ④

$$\boxed{\tan^2\theta = \frac{h_1}{h_2}}$$

Time of flight

$T_1 = \frac{2u\sin\theta}{g}$ — ⑤

and $T_2 = \frac{2u\cos\theta}{g}$ — ⑥

From equation ⑤ and ⑥

$$\boxed{\tan\theta = \frac{T_1}{T_2}}$$

and $$\boxed{R = 4\sqrt{h_1 h_2}}$$

and $$\boxed{R = \frac{1}{2} g T_1 T_2}$$

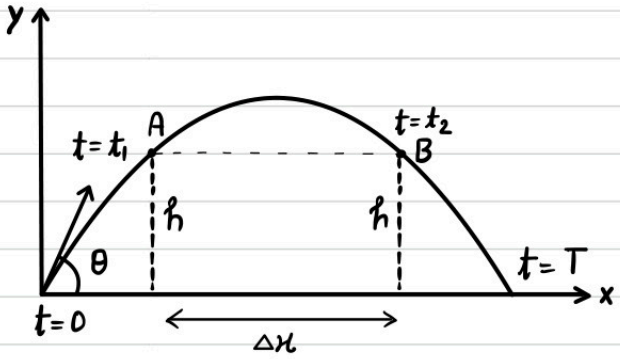
When two points A and B are at the same height :-

Time of flight

$T = t_1 + t_2 = \frac{2u\sin\theta}{g}$

Height

$h = \frac{1}{2} g t_1 t_2$

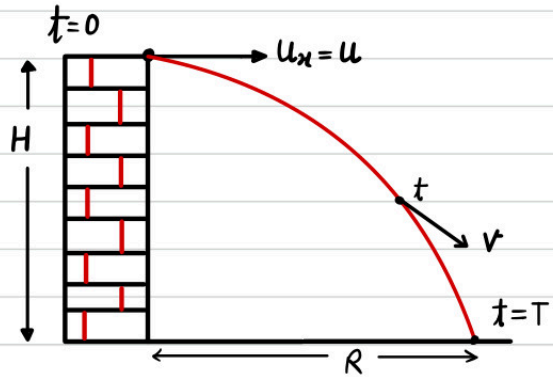


Average velocity b/w A and B $= u\cos\theta$

Horizontal separation b/w A and B is

$AB = \Delta x = u\cos\theta (t_2 - t_1)$

Projectile from a height :-



Here $u_x = u, \; a_x = 0$

$u_y = 0, \; a_y = -g$

After time t

$$y = u_y t + \frac{1}{2} a_y t^2$$

$$\Rightarrow \; -h = 0 \times t - \frac{1}{2} g t^2$$

$$\boxed{t = \sqrt{\frac{2h}{g}}}$$

Time of flight

$$T = \sqrt{\frac{2H}{g}}$$

Horizontal range

$$R = u_x \times T$$

$$\boxed{R = u\sqrt{\frac{2H}{g}}}$$

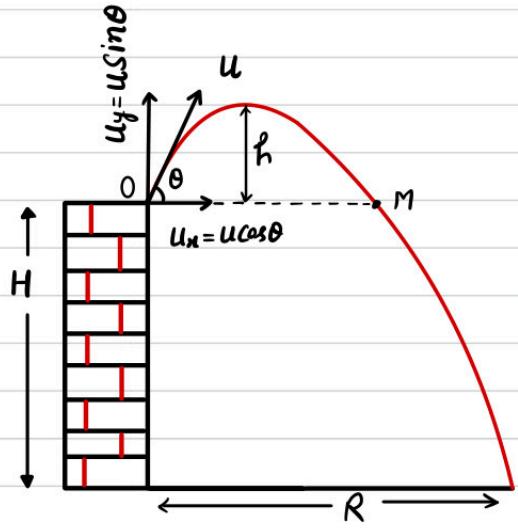
Equation of trajectory

$$x = ut \Rightarrow t = \frac{x}{u}$$

and

$$y = \frac{1}{2} g t^2$$

$$\boxed{y = \left(\frac{g}{2u^2}\right) x^2}$$



in diagram

$$h = \frac{u^2 \sin^2\theta}{2g}$$

displacement

$$OM = \frac{u^2 \sin 2\theta}{g}$$

and time

$$t_{OM} = \frac{2u\sin\theta}{g}$$

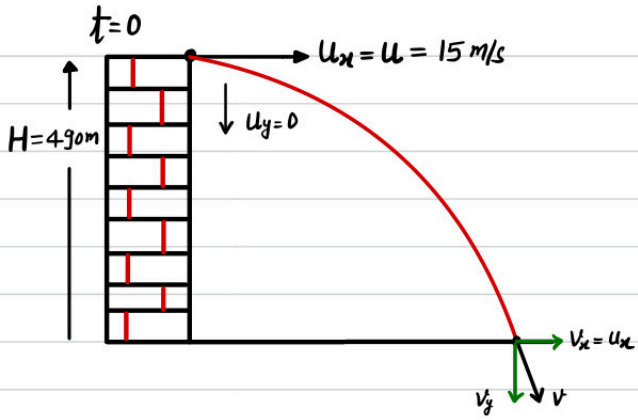
Total vertical displacement

$$\boxed{H = -u\sin\theta \times T + \frac{1}{2} g T^2}$$

Horizontal range

$$\boxed{R = u\cos\theta \times T}$$

Example 4.8 A hiker stands on the edge of a cliff 490 m above the ground and throws a stone horizontally with an initial speed of 15 m s^{-1}. Neglecting air resistance, find the time taken by the stone to reach the ground, and the speed with which it hits the ground. (Take $g = 9.8$ m s^{-2}).



Sol:- calculation time of flight

$$-H = u_y T - \frac{1}{2} g T^2$$

$$\Rightarrow T^2 = \frac{2H}{g} = \frac{2 \times 490}{9.8} = 4.9$$

$$T^2 = 100$$

$$T = 10 \text{ sec}$$

and

$$v_y = u_y - gT$$

$$v_y = -9.8 \times 10 = -98 \text{ m/s}$$

Resultant speed

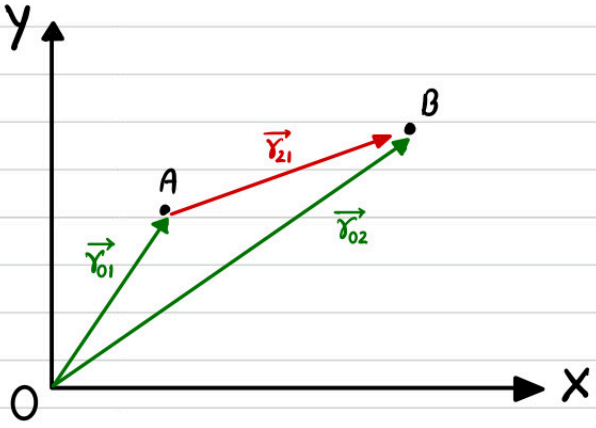
$$v = \sqrt{v_x^2 + v_y^2}$$

$$v = \sqrt{(98)^2 + (15)^2}$$

$$v = 99.14 \text{ m/s}$$

Relative motion

The positions of two moving particles A and B with respect to origin O at any instant are $\vec{r}_{01}$ and $\vec{r}_{02}$ respectively.



The position of B w.r.t to A is

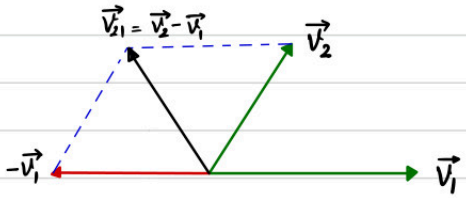
$$\vec{r}_{21} = \vec{r}_{02} - \vec{r}_{01}$$

differentiate with respect to time

$$\frac{d}{dt}(\vec{r}_{21}) = \frac{d}{dt}(\vec{r}_{02}) - \frac{d}{dt}(\vec{r}_{01})$$

$$\vec{v}_{21} = \vec{v}_2 - \vec{v}_1$$

Relative velocity of B w.r.t to A



Similarly $\vec{v}_{12} = \vec{v}_1 - \vec{v}_2$ Relative velocity of A w.r.t to B

Similarly relative acceleration

$\vec{a}_{21} = \vec{a}_2 - \vec{a}_1$ Relative acceleration of B w.r.t to A.

$\vec{a}_{12} = \vec{a}_1 - \vec{a}_2$ Relative acceleration of A w.r.t to B.

In one dimension motion

① If particles move in the same direction

$V_{AB} = V_A - V_B$

$V_{BA} = V_B - V_A$



② If particles move in opposite direction

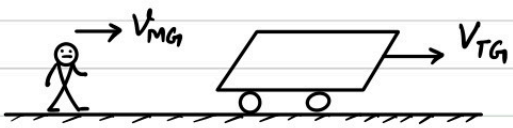
$V_{AB} = V_A - (-V_B)$

$V_{AB} = V_A + V_B$

$V_{BA} = -V_B - V_A = -(V_A + V_B)$



Example:-



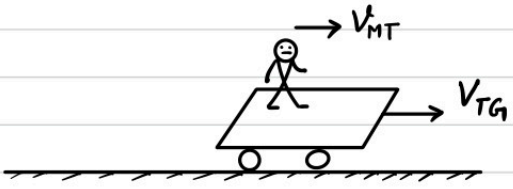
Relative velocity of trolley w.r.t to man

$V_{TM} = V_{TG} - V_{MG}$

or

$V_{TG} = V_{TM} + V_{MG}$

Relative velocity of man w.r.t to ground.

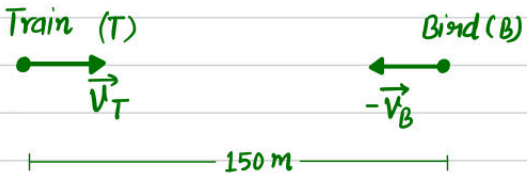


$V_{MT} = V_{MG} - V_{TG}$

$V_{MG} = V_{MT} + V_{TG}$

Question:- A 150 meter long train is moving towards the north with a speed of 10 m/s. A bird is flying towards the south parallel to the track with a speed of 5 m/s. How much time will it take for the bird to cross the train?

Sol:- The train and the bird are moving in opposite direction.

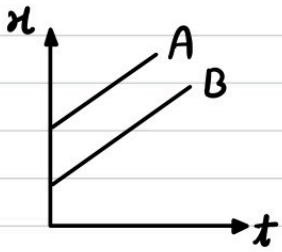


Thus, the relative velocity of bird w.r.t. to train

$$V_{BT} = 5 + 10 = 15 \text{ m/s}$$

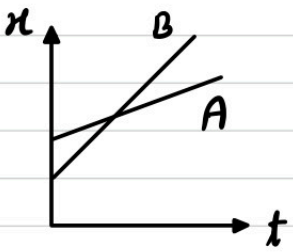
Time taken to cross the train $t = \frac{d}{V_{BT}} = \frac{150}{15} = 10 \text{ sec}$

Relative velocity from x-t graph:-

① 

Here $V_A = V_B$

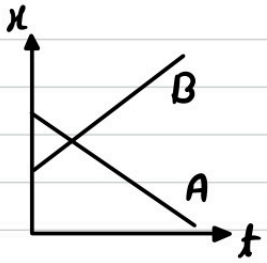
Thus $V_{AB} = V_{BA} = 0$

② 

Here $V_B > V_A$

Thus $V_{BA} = V_B - V_A = \oplus ve$

and $V_{AB} = V_A - V_B = \ominus ve$

③ 

Here $V_B = \oplus ve$

$V_A = \ominus ve$

Thus $V_{BA} = V_B + V_A \quad \oplus ve$

Question:- The velocities of particles A and B are $\vec{V}_A = (2\hat{i} + 4\hat{j})$ m/s and $\vec{V}_B = (3\hat{i} - 7\hat{j})$ m/s respectively. What will be the velocity of B as seen from A ?

Sol:-

Velocity of B as seen from A $\vec{V}_{BA} = \vec{V}_B - \vec{V}_A$

$$= 3\hat{i} - 7\hat{j} - 2\hat{i} - 4\hat{j}$$

$$= (\hat{i} - 11\hat{j}) \text{ m/s}$$

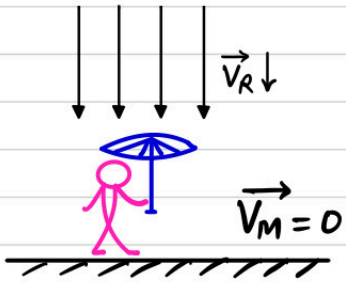
Speed $V_{BA} = \sqrt{1^2 + (-11)^2} = \sqrt{1 + 121}$

$$= \sqrt{122} \text{ m/s}$$

direction from X-axis $\tan\theta = \left(\frac{-11}{1}\right) \Rightarrow \theta = -\tan(11)$

Rain and umbrella:- Direction of umbrella → Rain velocity relative to person.

① Person is stationary:-

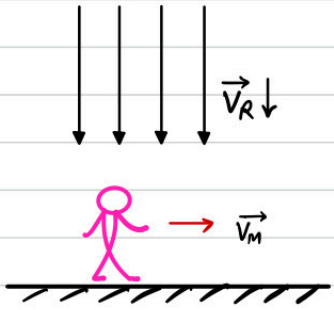


The umbrella should be held vertically.

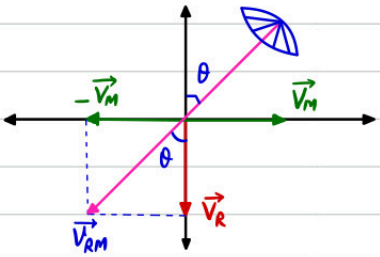
$$\vec{V}_{RM} = \vec{V}_R - \vec{V}_M \quad (\vec{V}_M = 0)$$

$$\vec{V}_{RM} = \vec{V}_R$$

② Rain is vertical, person is moving:-



Relative velocity of rain w.r. to person



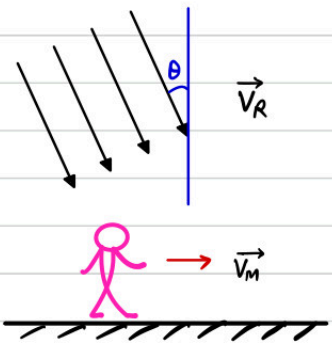
$$\vec{V}_{RM} = \vec{V}_R - \vec{V}_M$$

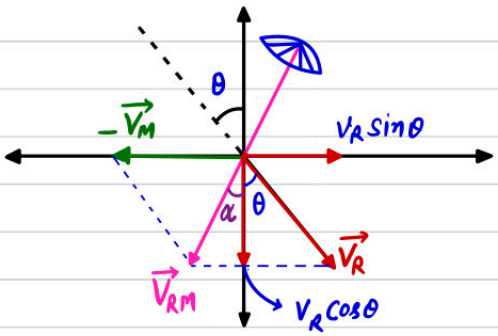
$$V_{RM} = \sqrt{V_R^2 + V_M^2}$$

direction of umbrella from vertical.

$$\theta = \tan^{-1}\left(\frac{V_M}{V_R}\right)$$

③ Rain falls at an angle, person is moving:-





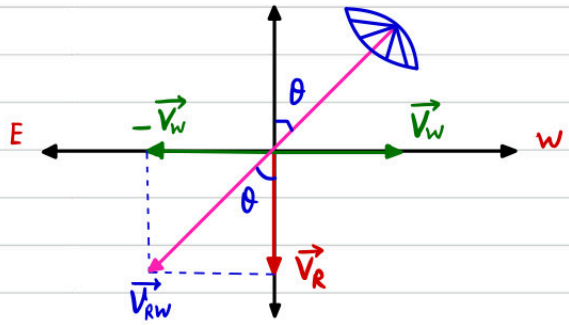
Relative velocity of rain $V_{RM} = \sqrt{(V_M - V_R \sin\theta)^2 + (V_R \cos\theta)^2}$

$$\tan\alpha = \frac{V_M - V_R \sin\theta}{V_R \cos\theta}$$

if $V_M = V_R \sin\theta$ then

$V_{RM} = V_R \cos\theta$ ⇝ Rain will appear to fall vertically.

Example 4.6 Rain is falling vertically with a speed of 35 m s^{-1}. A woman rides a bicycle with a speed of 12 m s^{-1} in east to west direction. What is the direction in which she should hold her umbrella ?



Sol:

Direction of umbrella

$$\theta = \tan^{-1}\left(\frac{V_w}{V_R}\right)$$

$$\theta = \tan^{-1}\left(\frac{12}{35}\right) \simeq 19^\circ$$ vertical to west

A man is walking on a horizontal road at velocity of 5 km/h and the rain appears to be falling vertically at 5 km/h to him. The magnitude of velocity of rain with respect to the ground is

(1) 5 km/h (2) $5\sqrt{2}$ km/h

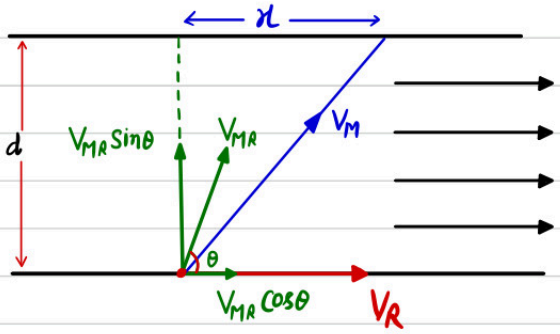
(3) 10 km/h (4) 4 km/h

River - Boat (swimmer) problem :-

V_{MR} = Velocity of man/boat relative to river (speed in still water)

V_R = Velocity of river

V_M = Velocity of man/boat relative to ground.



$$V_\perp = V_{MR}\sin\theta$$

$$V_{||} = V_{MR}\cos\theta + V_R$$

time taken to cross the river

$$t = \frac{d}{V_{MR}\sin\theta} = \frac{d}{V_\perp}$$

drift

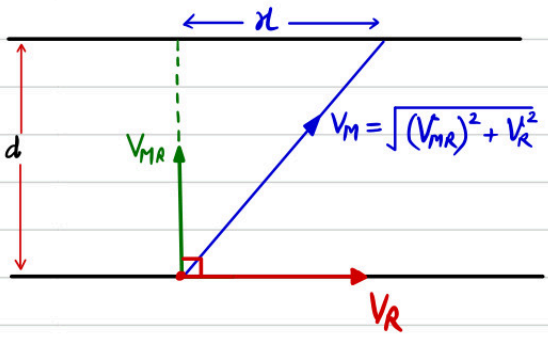
$$x = (V_{MR}\cos\theta + V_R)\times t = V_{MR}\cos\theta \times t + V_R \times t$$

$V_{MR}\cos\theta \times t$: due to man effort

$V_R \times t$: due to river flow

Minimum time to cross river

$$t = \frac{d}{V_{MR}\sin\theta}$$



if $V_\perp = V_{MR}\sin\theta$ is maximum then

$(V_\perp)_{max} = V_{RM}$ Thus $\sin\theta = 1 \Rightarrow \theta = 90^\circ$

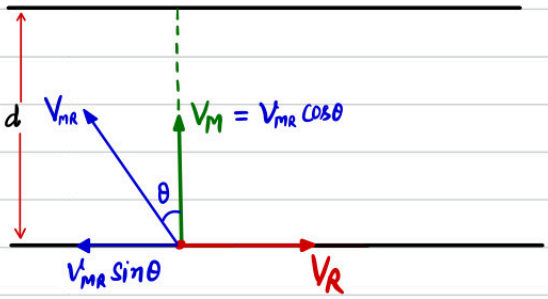
$$t_{min} = \frac{d}{V_{MR}}$$

drift $\quad x = V_R \times t = \frac{d \times V_R}{V_{MR}}$

Minimum distance to cross river (zero drift)

if drift is zero then

$$V_{\parallel} = 0 \Rightarrow V_R = V_{MR}\sin\theta \Rightarrow \sin\theta = \frac{V_R}{V_{MR}}$$



time taken to cross the river

$$t = \frac{d}{V_M} = \frac{d}{V_{MR}\cos\theta}$$

$$V_M = \sqrt{V_{MR}^2 - V_R^2}$$

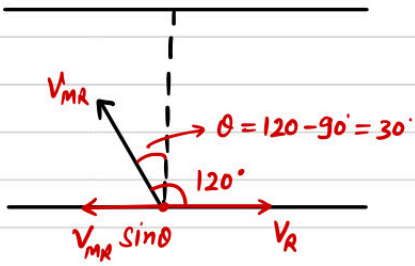
Question:- A person crosses a river in the shortest path. He starts swimming with a speed 4 km/h at an angle of 120° with the flow. Find the speed of river flow.

Sol:-

To cross the river in the shortest path

$V_{MR}\sin\theta = V_R$

$\Rightarrow V_R = 4 \times \sin 30^\circ = 4 \times \frac{1}{2} = 2$ km/hr



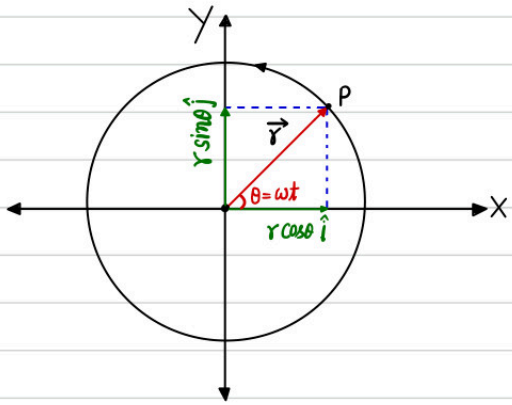
The speed of a swimmer in still water is 20 m/s. The speed of the river water is 10 m/s and it is flowing towards the east. If he is standing on the south bank and wants to cross the river along the shortest path, then at what angle with respect to the north direction should he make his swimming effort? That angle is: [NEET-2019]

(1) 30° West
(2) 0°
(3) 60°
(4) 45° West

Circular motion

When a particle moves on a circular path, its motion is called circular motion.
It is an example of 2-D motion.

Radius vector:- Vector showing the position from the centre of a particle moving on a circular path



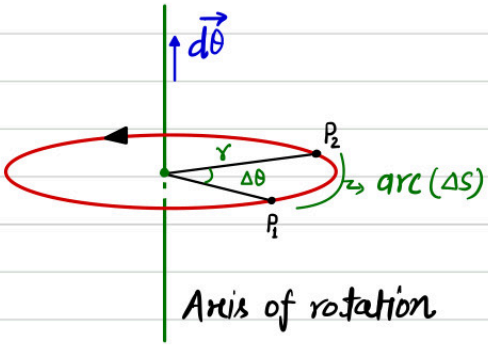
$$\vec{r} = r\cos\theta\,\hat{i} + r\sin\theta\,\hat{j}$$

$$\vec{r} = r\cos(\omega t)\,\hat{i} + r\sin(\omega t)\,\hat{j}$$

r → Radius

magnitude → constant, direction → variable

Angular displacement:- The angle swept by radius vector.



$$\text{Angular displacement} = \frac{\text{Arc}}{\text{Radius}}$$

$$\Delta\theta = \frac{\Delta s}{r}$$ → unit → rad (dimensionless)

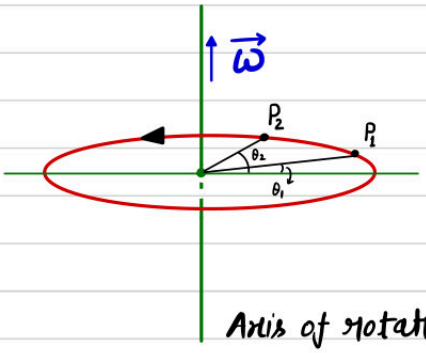
if $\Delta\theta \to 0$ then

Angular displacement = $\vec{d\theta}$ (Axial vector)

Angular velocity:- The rate of change of angular displacement.

Average angular velocity

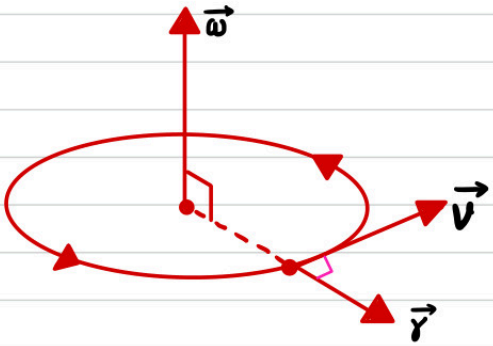
$$\omega_{av} = \frac{\Delta\theta}{\Delta t} = \frac{\theta_2 - \theta_1}{t_2 - t_1}$$ → rad/sec



instantaneous angular velocity

$$\vec{\omega} = \frac{d\vec{\theta}}{dt}$$

Relation b/w linear velocity and angular velocity :-



Angular disp. $\Delta\theta = \frac{\Delta s}{r} \Rightarrow \Delta s = r \cdot \Delta\theta$

or

$$\frac{\Delta s}{\Delta t} = r \cdot \frac{\Delta\theta}{dt}$$

if $\Delta t \to 0$ then

$$\frac{ds}{dt} = r\frac{d\theta}{dt} \Rightarrow \boxed{v = r\omega}$$

$\vec{v} \perp \vec{\omega} \perp \vec{r}$

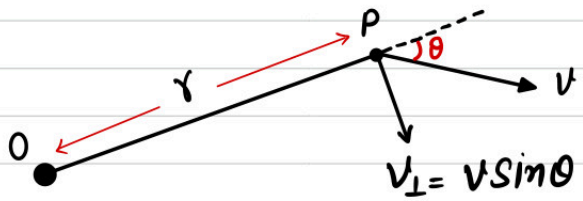
$\vec{v}\cdot\vec{\omega} = \vec{\omega}\cdot\vec{r} = \vec{v}\cdot\vec{r} = 0$

vector form $\boxed{\vec{v} = \vec{\omega} \times \vec{r}}$

Angular velocity of a particle w.r. to a point :-

$$\boxed{\omega = \frac{v_\perp}{r}}$$

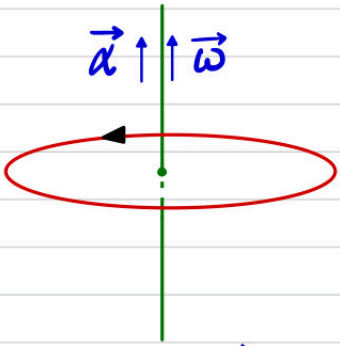
$$\boxed{\omega_{op} = \frac{v\sin\theta}{r}}$$



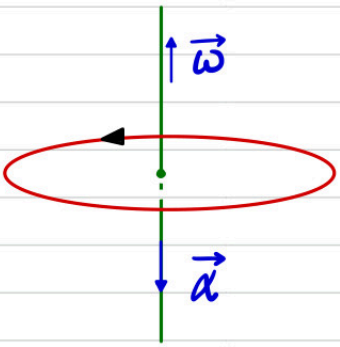
Angular acceleration :- The rate of change of angular velocity.

$$\alpha_{av} = \frac{\Delta\omega}{\Delta t} = \frac{\omega_2 - \omega_1}{t_2 - t_1} \rightsquigarrow \text{rad/sec}^2$$

instantaneous $\boxed{\vec{\alpha} = \frac{d\vec{\omega}}{dt}}$

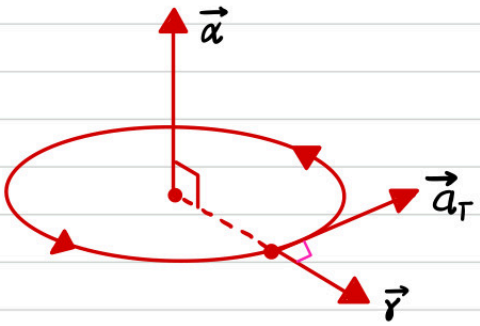


$\omega \uparrow$es



$\omega \downarrow$es

Relation b/w linear acceleration and angular acceleration :-



We know that

$$v = r\omega$$

after differentiation

$$\frac{dv}{dt} = r\frac{d\omega}{dt}$$

$$a_T = r\alpha$$

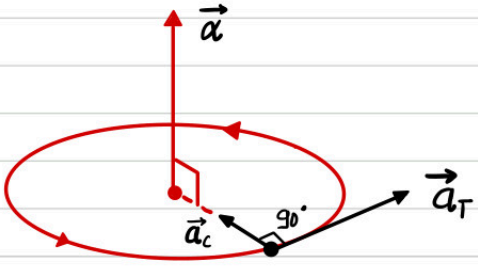
Vector form

$$\vec{a}_T = \vec{\alpha} \times \vec{r}$$

$\vec{a}_T \perp \vec{\alpha} \perp \vec{r}$

$$\vec{a}_T \cdot \vec{\alpha} = \vec{\alpha} \cdot \vec{r} = \vec{a}_T \cdot \vec{r} = 0$$

Acceleration in circular motion :-



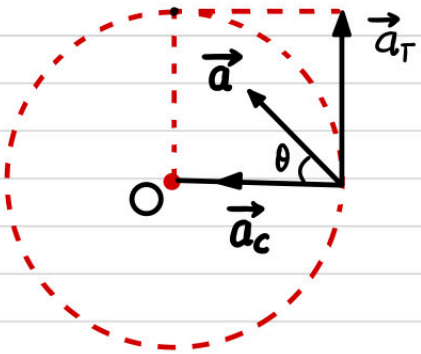
$$\vec{v} = \vec{\omega} \times \vec{r}$$

differentiate w.r.t. to time

$$\frac{d\vec{v}}{dt} = \vec{\omega} \times \frac{d\vec{r}}{dt} + \frac{d\vec{\omega}}{dt} \times \vec{r}$$

$$\vec{a} = (\vec{\omega} \times \vec{v}) + (\vec{\alpha} \times \vec{r})$$

$$\vec{a} = \vec{a}_c + \vec{a}_T$$



Here $\vec{a}_c = \vec{\omega} \times \vec{v}$ centripetal acceleration ↳ toward the centre

magnitude $a_c = \omega v = \omega^2 R = \frac{v^2}{R}$

$\vec{a}_T = \vec{\alpha} \times \vec{r}$ tangential acceleration

and $\vec{a}_c \perp \vec{a}_T$ $\quad \vec{a}_c \cdot \vec{a}_T = 0$

magnitude of resultant

$$a = \sqrt{a_c^2 + a_T^2}$$

$$\theta = \tan^{-1}\left(\frac{a_T}{a_c}\right)$$

Time period :- Time taken by a particle to complete one revolution.

time period $T = \frac{\text{circumference}}{\text{Linear speed}} = \frac{2\pi}{\text{Angular speed}}$

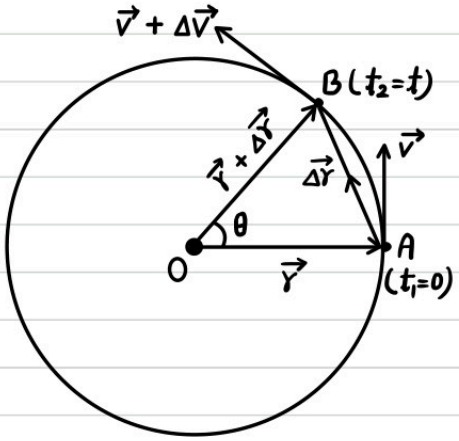
$$\boxed{T = \frac{2\pi r}{v} = \frac{2\pi}{\omega}} \rightarrow \text{unit} \rightarrow \text{second}$$

Frequency :- Number of revolution completed in one second. (n or f or ν)

$$\text{Frequency} = \frac{1}{\text{time period}} \Rightarrow \boxed{\nu = \frac{1}{T} = \frac{\omega}{2\pi}}$$

Uniform circular motion :- When a particle performs circular motion with constant speed, then this is called uniform circular motion of the particle.

in this motion, speed $|\vec{v}|$ = Constant



rate of change of speed $\frac{d|\vec{v}|}{dt} = a_T = 0$ and $\alpha = 0$

Angular velocity ω = Constant

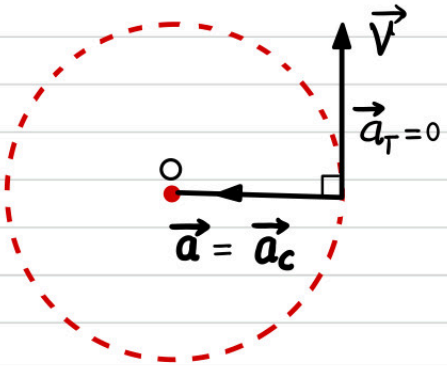
Change in magnitude of velocity $\Delta|\vec{v}| = 0$

magnitude of change in velocity $|\Delta\vec{v}| = 2v\sin\left(\frac{\theta}{2}\right)$

magnitude of displacement $|\Delta\vec{r}| = 2r\sin\left(\frac{\theta}{2}\right)$

magnitude of av. velocity $|\vec{v}_{av}| = \frac{2r\sin\left(\frac{\theta}{2}\right)}{t}$

Magnitude of av. acc^n $|\vec{a}_{av}| = \frac{2v\sin\left(\frac{\theta}{2}\right)}{t}$



resultant acc^n. $\vec{a} = \vec{a}_c + \vec{a}_T$ ($\vec{a}_T = 0$)

$$\boxed{\vec{a} = \vec{a}_c = \frac{v^2}{r}(-\hat{r}) = \omega^2 r(-\hat{r})}$$

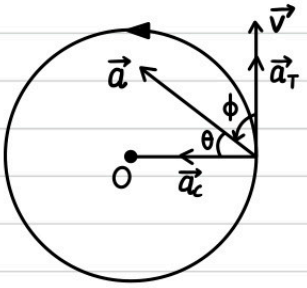
here $\vec{v} \perp \vec{a}$

Non uniform circular motion

(Speed change at each instant)

Speed increases

($\vec{v}$ and $\vec{a}_T$ are in same direction)



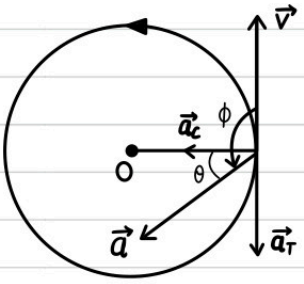
Magnitude of tan. acc^n. $a_T = \frac{d}{dt}|\vec{v}|$

Magnitude of resultant acc^n. $a = \sqrt{a_c^2 + a_T^2}$

Direction of resultant acc^n. with velocity vector $\phi = \left(\frac{\pi}{2} - \theta\right)$

Speed decreases

($\vec{v}$ and $\vec{a}_T$ are in opposite direction)



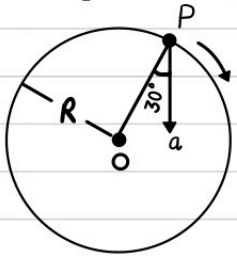
Magnitude of tan. acc^n. $a_T = \frac{d}{dt}|\vec{v}|$

Magnitude of resultant acc^n. $a = \sqrt{a_c^2 + a_T^2}$

Direction of resultant acc^n. with velocity vector $\phi = \left(\frac{\pi}{2} + \theta\right)$

Q :- In the given figure, a = 15m / s^2 represents the total acceleration of a particle moving in the clockwise direction in a circle of radius R = 2.5m at a given instant of time. The speed of the particle is

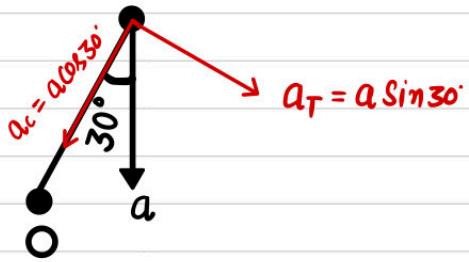
[NEET (Phase-2) 2016]



(1) 4.5 m/s
(2) 5.0 m/s
(3) 5.7 m/s
(4) 6.2 m/s

Solution:-

centripetal acceleration



$$a_c = \frac{v^2}{R}$$

$$v = \sqrt{a_c \times R}$$

$$v = \sqrt{a\cos 30° \times R}$$

$$v = \sqrt{15 \times \frac{\sqrt{3}}{2} \times 2.5} = \sqrt{18.75 \times 1.732} = \sqrt{32.475} \simeq 5.7 \text{ m/s}$$